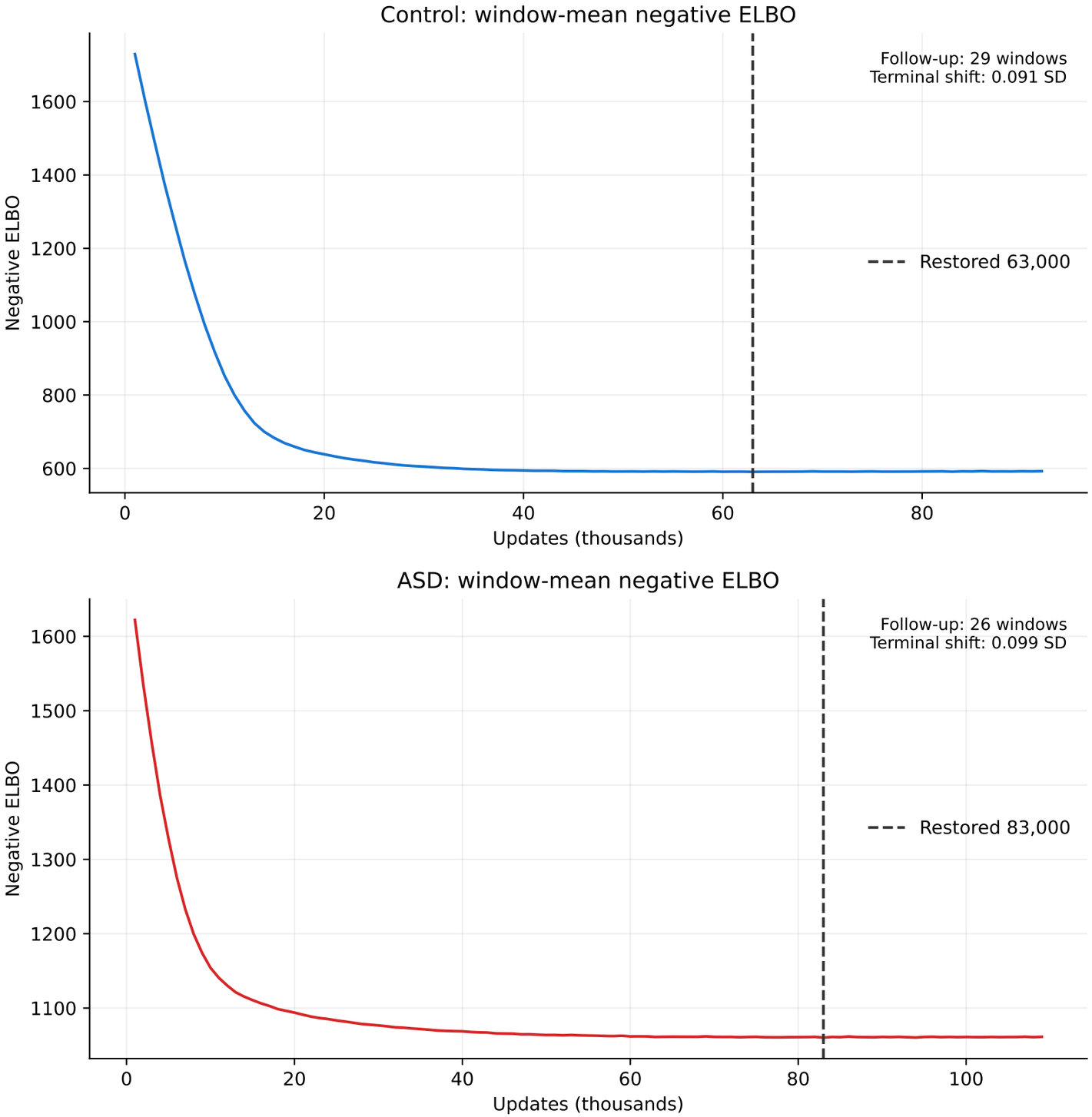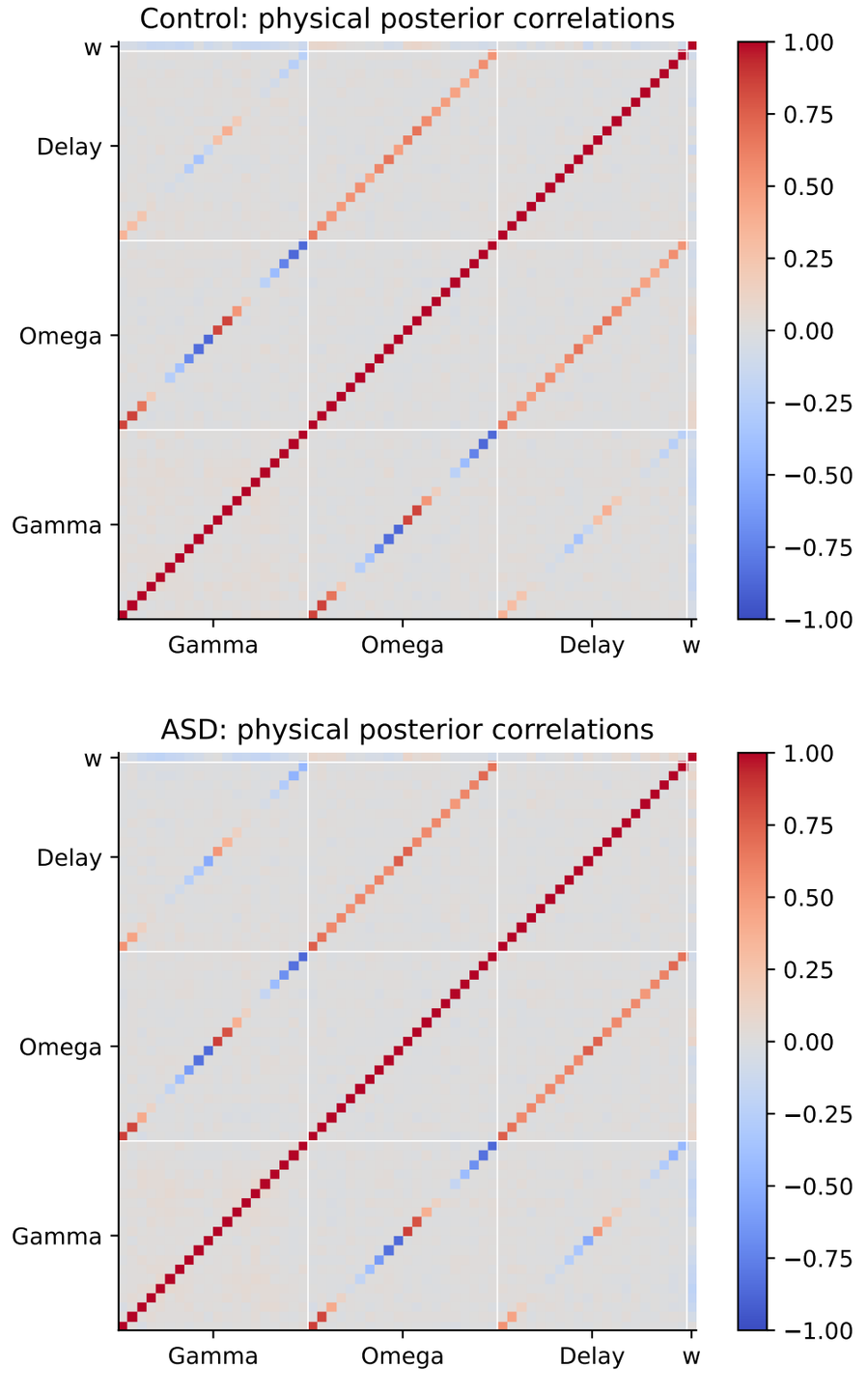Gamma–Omega SVI convergence and posterior dependence